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Sustainable Production of Dimethyl Ether from Syngas: Advancing CO₂ Conversion to Oilfield Chemicals

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Abstract

This study aims to design a sustainable and efficient process for the production of dimethyl ether (DME) from syngas, targeting its application as an upstream oilfield chemical. In addition to its growing role as a clean fuel alternative, DME is gaining increasing recognition as a highly effective solvent for subsurface applications. These include the removal of water and condensate blockages, improved performance in CO₂-enhanced oil recovery (CO₂-EOR), and solvent-assisted steam-assisted gravity drainage (SAGD). The proposed indirect route process comprises three main stages: syngas pre-treatment, methanol (MeOH) synthesis, and subsequent catalytic dehydration to DME. Process simulation was conducted using Aspen Plus, where reactor performance, separation efficiency, and material recycling were optimized to maximize yield. Thermodynamic model selection was central to the simulation accuracy, combining the Peng-Robinson (PR) equation of state with Wong-Sandler (WS) mixing rules for high-pressure systems, and the Non-Random Two-Liquid (NRTL) model for low-pressure, polar mixtures. Reactor models were selected based on known conversions, using RStoic modules for steady-state performance. The simulation successfully demonstrated that approximately 18,400 metric tons of 99.6% pure DME can be produced annually, based on a generic syngas feed rate of 100,000 metric tons per year. MeOH synthesis achieved a 36% conversion of CO and 20% conversion of CO₂, with a final MeOH purity of 99.5%, while subsequent dehydration yielded a 93% conversion to DME. DME purity and volatility were found to meet the requirements for multiple upstream applications. In particular, DME effectively dissolves water held in tight rock pore networks and clears condensate blockage via high miscibility with C₅⁺ hydrocarbons. For CO₂-EOR, blending 10-20 mol% DME with CO₂ significantly reduces minimum miscibility pressure and enhances crude oil mobilization. In solvent-aided SAGD, DME co-injection was shown to reduce the steam-oil ratio and shorten ramp-up time. These results support the conclusion that DME is not only technically feasible to produce from syngas at scale, but also strategically valuable for decarbonizing upstream oilfield operations.

Introduction

Over the past two decades, dimethyl ether (DME) has been recognized as an industrially significant and versatile chemical, increasingly valued within upstream oil and gas operations due to its unique

physicochemical properties (Chai et al. 2023). Its growth in industrial relevance arises from its potential as both an alternative fuel and an effective chemical solvent, highlighting its adaptability and integration within existing infrastructures (Chmielarz, 2024; Lotfollahzade and Hazlett, 2023). DME exhibits a simple molecular ether structure ($\text{CH}_3\text{-O-CH}_3$) characterized by low viscosity, high volatility, and a boiling point of -24°C (PubChem, 2024). These properties facilitate ease of handling and compatibility with current fuel-storage and distribution systems, critical for industrial-scale adoption. Furthermore, DME's high cetane number, typically ranging between 55 and 60, ensures efficient and rapid ignition in compression-ignition engines (Ying et al. 2008). This superior ignition capability contributes to enhanced engine performance, reduced combustion noise, and improved fuel efficiency, positioning DME as a promising alternative to traditional diesel fuels (Peinado et al. 2024; Ying et al. 2008). Importantly, DME combustion results in zero-sulfur emissions and significantly reduced particulate matter, aligning seamlessly with the rigorous international environmental standards aimed at limiting air pollution. In terms of safety and environmental impact, DME holds distinct advantages. It is noncarcinogenic, nonmutagenic, and presents minimal environmental hazards due to its clean combustion products, primarily carbon dioxide (CO_2) and water (Lotfollahzade and Hazlett, 2023; PubChem, 2024). These qualities significantly reduce health and safety concerns during handling, storage, and transportation, thus enhancing its overall viability as a sustainable industrial chemical and fuel.

The significance of DME at the present time is rooted in two converging drivers: tightening environmental regulations on conventional fossil fuels and the urgent need for alternative energy carriers that are both clean and readily integrated into existing systems. Whereas liquefied petroleum gas (LPG) and conventional diesel continue to dominate, their drawbacks, namely sulfur oxide emissions, particulate matter, and reliance on crude-oil derivatives, have become increasingly untenable (Ying et al. 2008). DME, by contrast, is produced from a variety of feedstocks, including syngas derived from natural gas, coal, biomass, and even captured CO_2 via indirect synthesis routes (Peinado et al. 2024; Chmielarz, 2024). This versatility in raw materials highlights its role as a bridge between current hydrocarbon-based infrastructures and future, lower-carbon energy systems. Consequently, DME's dual capacity as a fuel and as a chemical intermediate has fostered a broad spectrum of research and pilot deployments worldwide.

DME production is conventionally categorized into three primary methodologies: indirect synthesis, direct synthesis, and reactive distillation (Chmielarz, 2024). The indirect synthesis route involves the catalytic dehydration of methanol (MeOH) intermediate, leveraging extensively developed MeOH synthesis technologies already well-established in industry (Peinado et al. 2024; Lotfollahzade and Hazlett, 2023). This pathway, benefiting from mature technological infrastructure, demonstrates consistently higher product yields and better overall scalability. Direct synthesis, by contrast, entails synthesizing DME directly from syngas in a single reactor, theoretically simplifying the overall process by reducing intermediate steps. However, this approach is generally obstructed by more complex catalyst requirements and lower product yields, limiting its industrial deployment (Chmielarz, 2024). Reactive distillation integrates chemical reaction and product separation simultaneously within a single apparatus, potentially reducing capital and operational costs, although practical implementation at commercial scales remains relatively limited due to process complexities (Chmielarz, 2024). Consequently, the indirect route remains the predominant choice for commercial production owing to its reliability, efficiency, and scalability. The efficiency of DME production critically depends upon the quality of the syngas feedstock, necessitating comprehensive pretreatment processes. Effective removal of contaminants such as hydrogen sulfide (H_2S) and CO_2 is paramount to prevent catalyst poisoning, corrosion, and subsequent operational inefficiencies. Additionally, the water-gas shift (WGS) reactor plays an essential role in adjusting the syngas composition by optimizing the ratio of carbon monoxide (CO) to hydrogen (H_2), crucial for maximizing downstream conversion efficiencies and ensuring stable reactor performance. Green production of DME significantly enhances its attractiveness as a sustainable chemical and energy resource. This environmentally conscious production

methodology incorporates renewable or otherwise low-carbon feedstocks such as biomass-derived syngas and CO₂ captured from industrial emissions. Such sustainable production practices effectively mitigate greenhouse gas emissions, simultaneously integrating renewable energy streams into existing fossil-based energy infrastructures, thus contributing directly to global decarbonization efforts.

DME has demonstrated notable efficacy in several targeted upstream oil and gas applications (Chai et al. 2023). Its capability to remove water and condensate blockages within pipelines significantly enhances flow assurance, preventing operational disruptions and reducing maintenance costs (Ganjdanesh, 2016). In CO₂-enhanced oil recovery (CO₂-EOR) operations, DME facilitates efficient extraction of hydrocarbons, improving yields while simultaneously reducing environmental footprints through lower greenhouse gas emissions (Lee et al. 2021; Wang et al. 2024). Furthermore, its utility in steam-assisted gravity drainage (SAGD) processes as a solvent effectively reduces the viscosity of heavy crude oils, increasing extraction efficiency, reducing steam usage, and enhancing overall resource recovery (Chai et al. 2023). These practical advantages highlight the broad applicability and strategic importance of DME within contemporary upstream oil and gas operations. As illustrated in Figure 1, DME's role as an oilfield chemical extends across multiple upstream applications, positioning it as a versatile solution across various upstream applications.

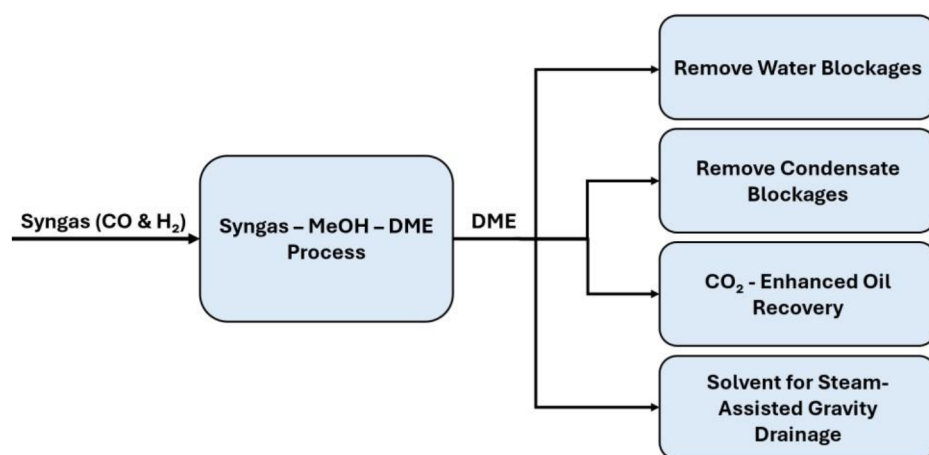


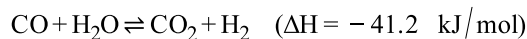
Figure 1—Upstream Applications of DME Derived from the Syngas-MeOH-DME Process

Process Methodology & Modeling

This section provides a comprehensive description of the integrated Syngas-to-MeOH-to-DME production process. The process is systematically segmented into three primary stages: syngas pre-treatment, MeOH synthesis and purification, and final DME synthesis. The indirect route was selected, as detailed previously, based on its proven commercial viability, efficiency, robust scalability, and established compatibility with existing MeOH synthesis infrastructures. For the simulation, an annual feed rate of approximately 100,000 metric tons of generic syngas, translating into around 12,030 kg/hr, was adopted.

The initial stage involves the pre-treatment of raw syngas to ensure compatibility and efficiency for downstream reactions. Raw syngas enters at ambient atmospheric pressure and high temperature, initially directed through a heat exchanger designed to cool the gas significantly. The cooling step induces partial condensation of water vapor, effectively enhancing downstream processes by reducing moisture levels. Subsequently, the resulting two-phase mixture is introduced into a flash drum, enabling efficient phase separation. Condensed water is effectively removed from the system as wastewater, leaving behind a vapor-rich syngas stream that continues onward. The vapor-rich syngas proceeds to an amine absorption column specifically configured to remove H₂S and CO₂. Methyl Diethanolamine (MDEA) acts as the selective solvent, effectively capturing these acid gases (Miller, 2011). Post-absorption, the amine solution rich in H₂S and CO₂ is routed to a stripping column, where regeneration occurs through thermal stripping. Heat appli

cation facilitates the breakdown of chemical bonds between the solvent and absorbed gases, regenerating lean MDEA solution suitable for recycling back to the absorption column (Addington and Ness, 2010). Acid gases separated during regeneration undergo subsequent cooling and condensation, ensuring complete separation and readiness for further handling, storage, or utilization elsewhere. Following amine scrubbing, the now purified syngas undergoes further compression to elevate its pressure to required operating levels. Integrated heat exchangers enhance energy recovery, lowering the gas temperature efficiently, thus optimizing thermal management across the process. The compressed and cooled syngas subsequently enters a WGS reactor. The primary function of this reactor is to shift the equilibrium of the syngas composition, converting CO and steam into additional H₂ and CO₂. The reaction proceeds as:



This mildly exothermic reaction is facilitated by a catalyst typically composed of iron oxide (Fe₃O₄) promoted with chromium oxide (Gr₂O₃). The reactor achieves a CO conversion of approximately 20.5%, effectively adjusting the H₂:CO ratio for optimal MeOH synthesis (Dasappa and Shivapuji, 2022). Following the WGS reaction, the syngas stream undergoes cooling and subsequent flash separation. This step effectively removes additional water formed during the shift reaction, resulting in further purification. Post water removal, the syngas enters a second acid gas removal stage employing the Selexol process. In this absorption column, Dimethyl Ethers of Polyethylene Glycol (DEPG) serve as the solvent, selectively capturing excess CO₂. Regeneration of the CO₂-rich solvent occurs similarly to the MDEA regeneration process, where recovered CO₂ is compressed and liquefied, providing potential for reuse in upstream applications (Miller, 2011; (Qayyum, Ali, and Ramazan, 2024). The configuration and sequence of units involved in this syngas pre-treatment stage are shown in Figures 2 and 3.

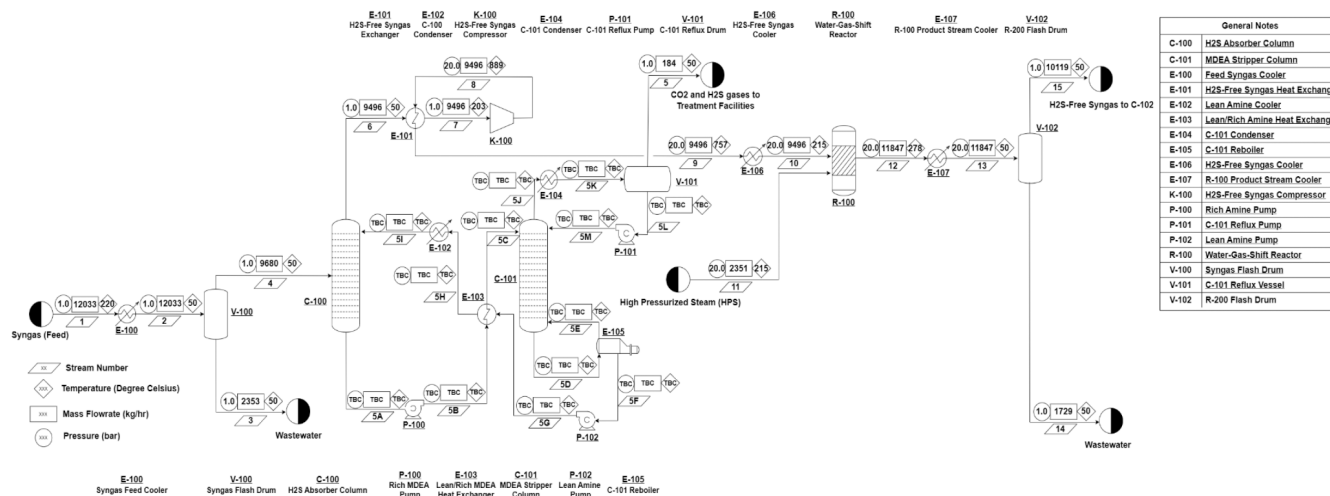


Figure 2—Syngas pre-treatment Section, sheet 1 out of 2 in the syngas-to-MeOH-to-DME production process.

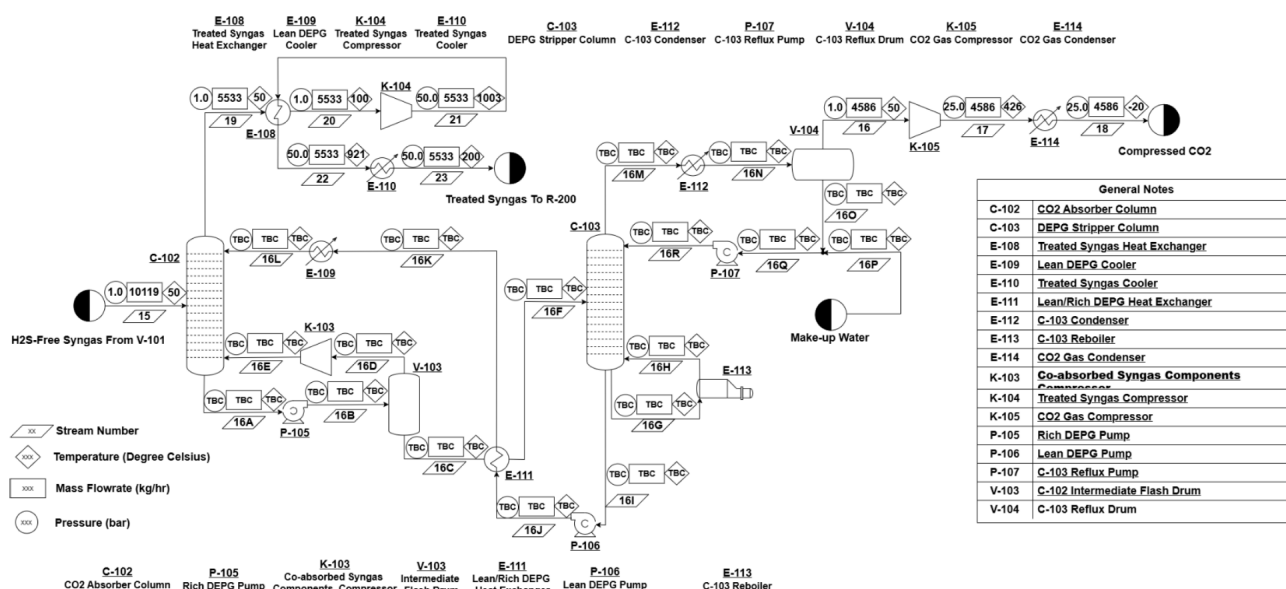
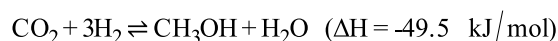


Figure 3—Syngas pre-treatment section, sheet 2 out of 2 in the syngas-to-MeOH-to-DME production process.

With syngas thoroughly treated and conditioned, the next stage of MeOH synthesis and purification commences. The refined syngas is compressed further to enhance reaction conditions, cooled, and then blended with recycled streams, creating a consistent feedstock mixture for the MeOH reactor. This combined syngas stream is introduced to an isothermal reactor specifically designed for MeOH synthesis, operating under controlled temperature and pressure. Within the reactor, catalytic hydrogenation of CO and CO₂ occurs via the following reactions:



These reactions are catalyzed using a copper-based system, typically comprising Cu/ZnO/Al₂O₃. The reactor achieves a CO conversion of approximately 36% (Bisotti et al. 2022), while CO₂ hydrogenation proceeds with a conversion of about 20% (Leonzio, 2018). The reactor effluent consists predominantly of MeOH along with unconverted syngas, water, and minor amounts of hydrocarbons. This mixture undergoes an expansion step, reducing its pressure and temperature, followed by cooling and entry into a flash separation unit. The flash drum effectively separates unreacted gases from the MeOH-rich liquid phase. The recovered gaseous phase is subsequently recompressed and largely recycled back to the MeOH reactor, optimizing production yields. However, to prevent the accumulation of inert gases within the recycling loop, a minor fraction of these gases is periodically purged from the system. The MeOH-rich liquid phase exiting the flash drum contains residual water, requiring further purification to meet the desired purity levels for downstream processes. Consequently, this mixture is introduced into a specifically configured distillation column. Optimized operating conditions within the column effectively separate MeOH as the overhead distillate product. This purified MeOH stream is then condensed, pressurized, and prepared for the subsequent synthesis stage of DME production. The separated water, comprising the heavier phase from the distillation column, is discharged from the process as wastewater. The configuration of the MeOH synthesis and purification section is illustrated in Figure 4.

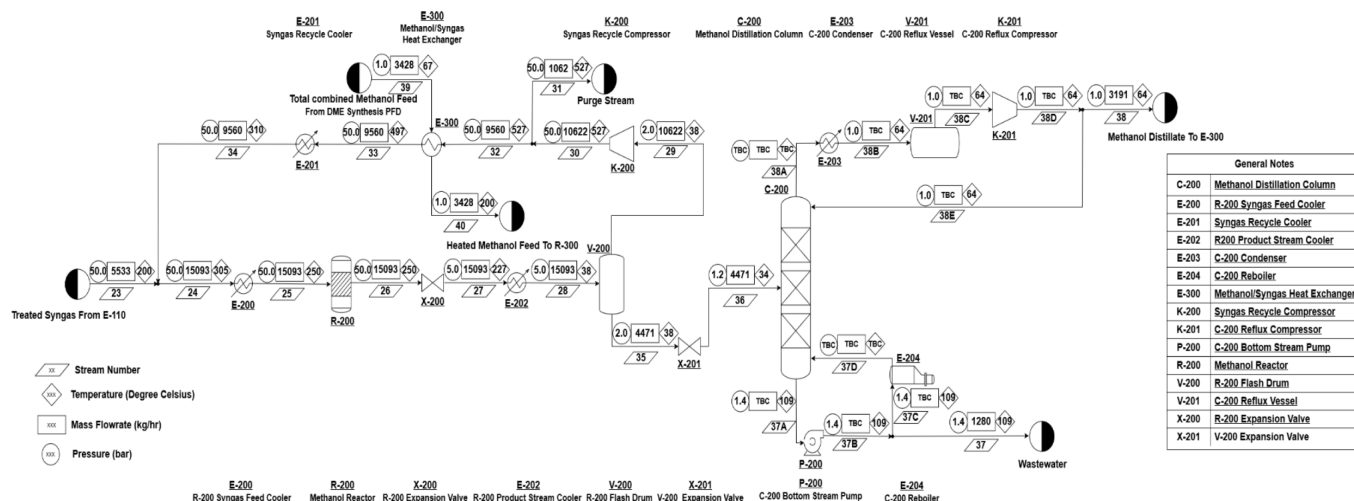
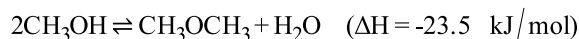


Figure 4—MeOH synthesis and purification section in the syngas-to-MeOH-to-DME production process.

The final stage of the integrated process involves synthesizing DME from purified MeOH. Initially, the purified MeOH is combined with recycled MeOH streams to ensure efficient utilization of recovered materials, enhancing overall production economics. This combined MeOH stream is preheated to optimal reaction conditions before introduction into the final reactor, specifically designed for MeOH dehydration. The reaction proceeds as:



This exothermic dehydration reaction is catalyzed by a solid acid catalyst, typically γ -alumina ($\gamma\text{-Al}_2\text{O}_3$), under adiabatic conditions. The reactor achieves a MeOH conversion of approximately 93% (Lotfollahzade and Hazlett, 2023). The resulting reactor effluent, consisting primarily of DME alongside unconverted MeOH and water, undergoes sequential compression and cooling steps, preparing the mixture adequately for the subsequent distillation stage. The distillation column designed for DME purification operates under optimized reflux conditions, ensuring high product purity. DME is recovered as the overhead distillate product. This purified DME product is suitable for various commercial and industrial applications, fulfilling market specifications. Remaining MeOH and water from the bottoms of the distillation column undergo further purification in an additional distillation step, recovering MeOH at a high efficiency. This recovered MeOH is recycled predominantly back into the MeOH feed stream, significantly enhancing process efficiency and sustainability. To avoid accumulation, a minor portion is purged periodically. The final purified DME product, obtained as a liquid, is appropriately stored and made available for commercial distribution, concluding the integrated Syngas-to-MeOH-to-DME production route. The configuration of the DME synthesis section is illustrated in Figure 5.

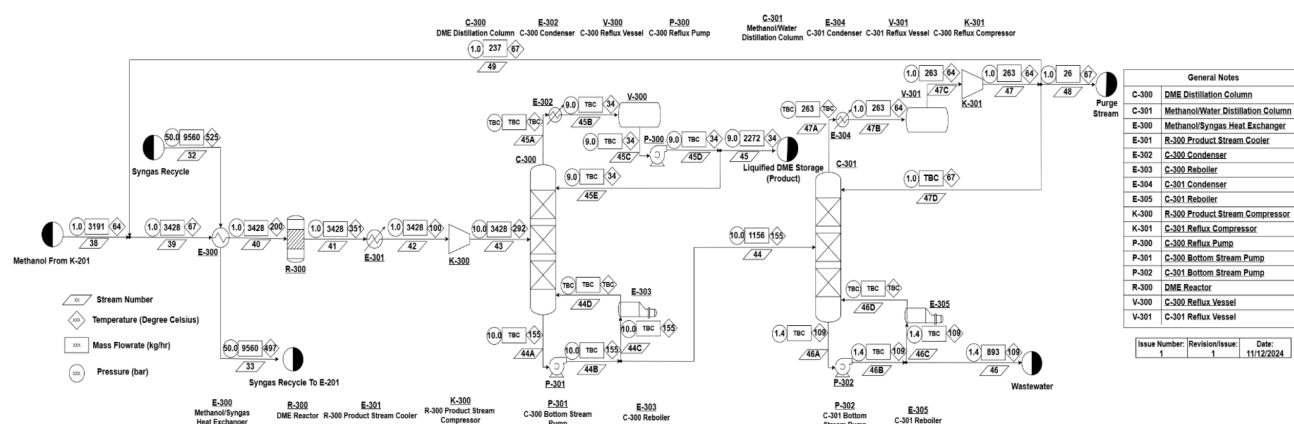


Figure 5—DME synthesis through the indirect route section in the syngas-to-MeOH-to-DME production process.

Simulation Decisions. In selecting appropriate thermodynamic models for the process simulation, two primary categories were considered: Equation of State (EOS) models and Activity Coefficient models. EOS models describe the behavior of fluids by mathematically relating key thermodynamic properties such as pressure, volume, and temperature (Comsol, 2019, Nielsen, 2020). EOS models also incorporate terms that account for molecular interactions and the finite size of molecules, often calibrated using experimental data. This makes EOS models particularly well-suited for simulating high-pressure systems. Activity Coefficient models, on the other hand, are designed to capture the non-ideal behavior of mixtures by calculating activity coefficients (Comsol, 2019, Nielsen, 2020). These models are especially accurate for systems involving polar components, such as water and MeOH, under low to moderate pressure conditions. However, their performance tends to deteriorate at higher pressures. This is because the parameters used to fit these models are usually derived from low- to medium-pressure data, and thus do not reliably account for the non-ideal behavior, fugacity corrections, or phase equilibrium at elevated pressures (Patel, 2023). Given that the methanol synthesis reactor (R-200) operates at pressures as high as 50 bar (Léonard, Giuliani, and Villarreal-Singer, 2016), an EOS model is more appropriate for modeling the reactor and other high-pressure units. EOS models offer better accuracy in such conditions due to their robust handling of pressure-dependent thermodynamic properties. Nevertheless, for units operating at lower pressures and involving strongly polar mixtures, such as MeOH-water distillation columns, Activity Coefficient models offer improved accuracy (Nielsen, 2020). Therefore, EOS models are used as the primary thermodynamic base throughout the simulation, while Activity Coefficient models are selectively applied to low-pressure, highly nonideal systems.

When evaluating which EOS model to use, a comparative study involving MeOH production from biomass-derived syngas found minimal differences between the Peng-Robinson (PR) and Soave-Redlich-Kwong (SRK) models. Both models exhibited similar activity correction factors and yielded nearly identical methanol molar fractions in simulation results. These findings are consistent with widespread industrial and academic use of both PR and SRK models in MeOH production, as both are known to provide reliable and accurate predictions (Adil, Shivapuji, and Rao, 2022). To enhance the predictive accuracy of EOS models for polar systems, they are often coupled with advanced mixing rules (Moioli and Pellegrini, 2016; Whitson, 2024). Incorporating binary interaction parameters through such rules improves the models' ability to represent vapor-liquid equilibria (VLE) under varying conditions, particularly in high-pressure and high-temperature environments. In this study, the PR model with Wong-Sandler (WS) mixing rules was selected as the primary thermodynamic base. This combination is especially effective for high-pressure systems exceeding 50 bar and offers enhanced accuracy in mixtures containing both polar components (e.g., MeOH and water) and non-polar species (e.g., CO₂ and H₂) (Calderon, 2006). The WS mixing rule was chosen over alternatives such as Huron-Vidal due to its superior performance in strongly non-ideal, polar systems (Hsieh

and Lin, 2012). For low-pressure units involving highly polar components, Activity Coefficient models are more appropriate. The Non-Random Two-Liquid (NRTL) model was selected for this purpose, as it is specifically developed to handle strongly non-ideal liquid-phase behavior. NRTL will be applied to model the first and third distillation columns, both of which are involved in separating MeOH and water at near-atmospheric pressures. Compared to PR-WS, the NRTL model offers more precise VLE predictions under these conditions, making it more suitable for use in DSTWU column simulations (Calderon, 2006; de Klerk and Schwarz, 2023), particularly when significant polar interactions are present.

While the NRTL model is well-suited for predicting liquid-phase behavior and VLE in low-pressure systems such as distillation columns, it lacks an equation of state and therefore cannot reliably handle vapor-phase thermodynamics, which are essential for compressor simulations. In contrast, the PR-WS model is more adept at modeling vapor-phase behavior but is less accurate in capturing the complexities of highly polar, low-pressure mixtures. To address this, a hybrid thermodynamic approach was employed: the NRTL model combined with the Redlich-Kwong equation of state (NRTL-RK). This method blends an activity coefficient model with an EOS model to extend the predictive accuracy across both liquid and vapor phases over a wider range of pressures than either model could achieve independently. The result is a model capable of accurately representing low- and moderate-pressure, non-ideal polar mixtures, as well as the vapor-phase corrections required for compressor operations. Additionally, it offers improved accuracy for polar components compared to PR-WS, making it particularly suitable for scenarios where both pressure and polarity vary significantly. When setting up the initial Aspen Plus simulation, special consideration was given to selecting the most appropriate reactor models. The choice involved balancing the trade-off between computational efficiency and thermodynamic accuracy. Three reactor model types were assessed: RStoic, REquil, and RGibbs.

The RStoic reactor is based on predefined stoichiometry and user-specified conversion rates. It offers the highest level of control over individual reactions and is computationally efficient, making it ideal for systems where reaction extents are known or need to be constrained. However, it does not account for equilibrium or kinetic behavior, limiting its usefulness in systems where conversion depends on reaction dynamics. The REquil model simulates reactions that reach thermodynamic equilibrium. It is particularly valuable for processes like the WGS reaction, where equilibrium dictates the final composition. Although it cannot simulate kinetic limitations, REquil provides a more realistic representation of systems where equilibrium is a valid assumption. The RGibbs reactor goes a step further by minimizing the system's Gibbs free energy to determine the equilibrium state. This model is advantageous for handling multiphase and multi-reaction systems without requiring detailed reaction mechanisms. It simplifies model setup and is highly flexible, but it is computationally intensive and offers the least control over individual reaction pathways. Moreover, it relies heavily on the accuracy of the thermodynamic data provided by the models used. Selecting the appropriate reactor model in Aspen Plus depends on the nature of the chemical system, the availability of reaction data, and the desired balance between control and realism. Each model has unique advantages and limitations that must be weighed according to the specific requirements of the simulation.

Although the WGS reaction involves equilibrium behavior, and might initially appear best suited for modeling with the REquil reactor, a different approach was taken in this simulation. Based on prior results from GAMS optimization, the expected extent of the WGS reaction is already known. This makes it feasible to treat the reaction using a stoichiometric model (Demirel and Ayas, 2017). Under the defined process conditions and in the presence of a catalyst, the conversion is assumed to remain stable at a steady state. With a known conversion rate, the WGS unit can be modeled using RStoic without sacrificing accuracy, which also simplifies the simulation. This approach prioritizes an accurate mass balance for the entire process rather than modeling the equilibrium behavior of a single reactor in detail. It is important to recognize the limitations of this simplification. If the reaction conditions no longer favor the forward WGS reaction, such as if the temperature deviates from the validated range or the catalyst deactivates over time, the actual conversion may fall below the assumed value. These potential inaccuracies represent the trade-off for using

a simpler stoichiometric model. However, given that the main objective of this preliminary study is to evaluate process feasibility and determine expected flow rates under steady-state conditions, the increased control provided by the RStoic model was deemed more beneficial than the minor accuracy gains from a full equilibrium-based model.

The RStoic reactor model was selected for both MeOH and DME synthesis units, as the reactions involved can be reliably represented using fixed stoichiometry and known conversions (Salahudeen et al. 2024), which are 36% for MeOH and 93% for DME. These values are well-established in literature and align with the process conditions used in this study. The RGibbs model was deemed unsuitable because it can introduce unrealistic side reactions that are unlikely to occur in the presence of selective catalysts, which are designed to suppress such alternate pathways. Likewise, the REquil reactor was not used, as it assumes the reaction proceeds to full thermodynamic equilibrium, an assumption that may not hold under actual operating conditions. Using RStoic offers several practical advantages. It simplifies reactor modeling by focusing on known conversions and stoichiometric balances, eliminating the need for complex thermodynamic or kinetic calculations. Also, its simplicity enhances convergence stability, reducing the likelihood of computational errors during simulation.

For the preliminary modeling of distillation units, the DSTWU model was used instead of the more rigorous RADFRAC model. DSTWU applies a shortcut method based on key assumptions, such as constant relative volatility and simplified stage calculations via the Fenske-Underwood-Gilliland (FUG) approach (Ray, 1995; Henley, Seader, and Roper, 2011). These simplifications drastically reduce computational demands, making DSTWU the ideal choice for early-stage design, where the primary goal is to obtain estimates of column performance. The thermodynamic model selection was based on pressure and mixture polarity. The NRTL model was applied to the MeOH-water separation columns, as it provides accurate VLE predictions for strongly non-ideal, polar mixtures at low pressures. For the DME-MeOH column, which operates at elevated pressures, the PR-WS was selected due to its enhanced reliability under high-pressure conditions and its ability to handle systems containing both polar and non-polar components.

Results & Discussion

The process simulation yielded several key results that inform the feasibility and efficiency of the proposed Syngas-to-MeOH-to-DME production route. Beginning with the MeOH synthesis stage, the reactor output stream demonstrated a mass flow rate of approximately 15,100 kg/hr. Within this stream, MeOH was produced with a composition of 36.0% by mass, signifying an effective conversion of syngas to methanol under the specified reactor conditions. This intermediate product was subsequently purified via distillation, achieving a methanol purity of 99.5%, making it suitable for use in the final synthesis stage.

In the DME synthesis stage, the purified methanol feed underwent catalytic dehydration, achieving a conversion rate of approximately 93%. This resulted in a final DME product stream with a mass flow rate of 2,270 kg/hr and a product purity of 99.6%. On an annual basis, this corresponds to a DME production rate of approximately 18,400 metric tons per year. This level of output aligns well with the process objectives and demonstrates the viability of the indirect synthesis route when simulated under realistic operating conditions. The use of appropriate thermodynamic models and reactor configurations, tailored to the pressure, composition, and phase behavior of each process segment, contributed to the accuracy and stability of the simulation results. Critically, the high-purity DME produced via this method meets the stringent requirements for oilfield chemical applications, particularly in upstream operations. Its low viscosity, volatility, and solvent properties make it highly effective for water and condensate blockage removal, CO₂-EOR, and viscosity reduction in SAGD, key advantages highlighted in earlier sections. The simulation thus confirms not only the technical feasibility of scaled DME production but also its practical readiness for integration into upstream hydrocarbon systems.

DME Upstream Oilfield Applications. With the indirect syngas route shown to produce high-purity DME suitable for field use, focus now shifts to its operational performance as an oilfield chemical. The following section explores the compound's effectiveness across four key upstream applications: removal of water and condensate blockages, enhanced oil recovery using CO₂-DME blends, and solvent-assisted SAGD.

Removal of Water Blockages. Water held by capillary forces in nano- to micro-scale pore throats can reduce gas relative permeability by an order of magnitude in tight reservoirs (Ganjdanesh et al., 2016). Injecting a 0.2 – 0.3 pore-volume slug of DME dissolves the trapped aqueous phase while simultaneously lowering water/hydrocarbon interfacial tension; numerical and core-flood studies showed that near-initial gas deliverability was restored whereas methanol or CO₂ treatments achieved only partial clean-up (Neshat et al., 2020). The high vapor pressure of DME accelerates flow-back after the treatment and avoids surfactant residue, limiting secondary formation damage (Ganjdanesh et al., 2016). Moreover, DME can be handled with existing LPG infrastructure, keeping surface modifications minimal.

Removal of Condensate Blockages. Retrograde condensation near the wellbore creates a liquidsaturation halo that chokes gas flow. DME is completely miscible with C₅⁺ hydrocarbons under reservoir conditions and exhibits a viscosity two orders of magnitude lower than the condensate. Single-cycle DME soaks dissolved up to 80% of the condensate bank in laboratory and simulation work, raising predicted gas productivity by 40-60% compared with the untreated case (Ganjdanesh et al., 2016; Neshat et al., 2020). After dissolution, the solvent flashes to the gas phase, sweeping the mobilized condensate towards the well and ensuring high chemical recovery for surface recycle. The treatment can be deployed in a batch or continuous mode using standard LPG pumps and cylinders with no additional corrosion risk.

CO₂-EOR. Adding 10-20 mol% DME to the injected CO₂ stream lowers the minimum miscibility pressure by up to 30% and cuts interfacial tension by nearly two orders of magnitude relative to pure CO₂ (Kong et al., 2021). Core-floods and PVT tests further show that a ternary CO₂-C₂H₆-DME mixture reduces crude-oil viscosity by 40% at 25 MPa and 60 °C, translating into a 15-20% incremental oil recovery over baseline CO₂ huff-n-puff cycles in tight cores (Wang et al., 2024). DME also increases the density and viscosity of the gas phase, mitigating channeling and improving sweep, while its partial water solubility suppresses asphaltene precipitation. Because DME partitions into both the oleic and gaseous phases, most of the solvent is produced back and can be fractionated for recycling with minimal make-up.

Solvent-Aided SAGD. Coinjecting 2-10 mol% DME with steam lowers the steam-oil ratio by 30-40% and accelerates bitumen mobilization relative to steam-only SAGD (Sheng et al., 2018). Physical-model experiments confirm that a 10 mol% DME addition lifted cumulative recovery by 20% and more than doubled the peak production rate while maintaining solvent recoveries above 80% (Li et al., 2023). DME dissolves in both water and bitumen, raising chamber-edge temperature and delivering latent-heat release at the oil interface, which together shorten the ramp-up period. Closed-loop separation at surface, enabled by DME's distinct boiling point (–25°C at 1 bar), minimizes solvent loss and emissions. Consequently, fuel use, water demand and associated CO₂ emissions decline, helping operators meet carbon-intensity targets without major equipment changes. In addition to its practical performance in upstream operations, the deployment of DME contributes meaningfully to CO₂ reduction across multiple fronts. Critically, the indirect synthesis route incorporates captured CO₂ into the syngas feed, enabling carbon reutilization rather than net emission. During MeOH synthesis, both CO and CO₂ are converted into intermediates, effectively transforming waste carbon into value-added products. This is particularly impactful when the syngas originates from biomass or flue gas, reducing the overall carbon footprint. In upstream applications, such as CO₂-EOR, SAGD, and blockage removal, DME further offsets emissions by enhancing recovery efficiency and displacing more carbon-intensive solvents and fuels. Its high volatility and miscibility promote effective recycling, and its combustion produces only CO₂ and water, avoiding particulate and sulfur oxide emissions. While CO₂ is ultimately released, much of it is derived from previously captured sources, thereby shifting the process toward net neutrality. Nevertheless, the full decarbonization potential of DME depends on closed-loop recovery, solvent recycling,

and sustainable syngas sourcing. Without these safeguards, the process risks undermining its environmental gains. Therefore, while DME holds substantial promise as a low-carbon oilfield chemical, its climate benefits are critically dependent on how comprehensively carbon is managed across the production and deployment cycle.

Conclusion

This study demonstrated the technical feasibility and industrial relevance of producing DME via the indirect Syngas-to-MeOH-to-DME route. Through rigorous process simulation, high-purity DME was successfully synthesized at a production rate of approximately 18,400 metric tons per year, with conversion efficiencies of 36% for MeOH and 93% for DME. The selected process route, supported by mature MeOH synthesis infrastructure, proved both scalable and operationally efficient. Thermodynamic model selection, primarily the PR with WS mixing rules and the NRTL model, ensured accurate representation of phase behavior across varying pressures and polarities. Additionally, the application of RStoic reactors provided a stable and computationally efficient simulation framework. The resulting DME meets the quality and performance requirements for critical upstream oilfield applications, including water and condensate blockage removal, CO₂-enhanced oil recovery, and solvent-assisted SAGD, while also presenting a viable pathway for CO₂ utilization and emissions reduction.

Reference

- Addington, L. and Ness, C. March 2010. An evaluation of general "rules of thumb" in amine sweetening unit design and operation, *Digitalrefining*, <https://www.digitalrefining.com/article/1000449/an-evaluation-of-general-rules-of-thumb-in-amine-sweetening-unit-design-and-operation> accessed 18 May 2025.
- Adil, A., Shivapuji, A., and Rao, L. 2022. Thermodynamic analysis for methanol synthesis using biomass-derived syngas. *Biomass Conversion and Biorefinery* **12** (5): 1819–1834. <https://doi.org/10.1007/s13399-022-02338-y>.
- Bisotti, F., Fedeli, M., Prifti, K. et al 2022. Impact of Kinetic Models on Methanol Synthesis Reactor Predictions: In Silico Assessment and Comparison with Industrial Data. *Industrial & Engineering Chemistry Research* **61** (5): 2206–2226. <https://doi.org/10.1021/acs.iecr.1c04476>.
- Calderon, S. 2006. Aspen Thermodynamic Property Models, University of Sao Paulo, <http://sites.poli.usp.br/d/pqi2408/physpropmodels.pdf> accessed 23 May 2025.
- Chai, M., Chen, Z., Nourozieh, H. et al 2023. Introduce dimethyl ether (DME) as a solvent for steam-assisted gravity drainage (SAGD) co-injection: An effective and environmental application. *Fuel* **341**: 127639. <https://doi.org/10.1016/j.fuel.2023.127639>.
- Chmielarz, L. 2024. Dehydration of Methanol to Dimethyl Ether—Current State and Perspectives. *Catalysts* **14** (5): <https://doi.org/10.3390/catal14050308>.
- Comsol Documentation 2019. Thermodynamic Models, https://doc.comsol.com/5.6/doc/com.comsol.help.chem/chem Ug thermo_11.21.html accessed 23 May 2025.
- Dasappa, S. and Shivapuji, A. 2022. Biomass gasification: Thermochemical route to energetic bio-chemicals. *Advanced Biofuel Technologies*, Elsevier,, (), 305–332. <https://doi.org/10.1016/B978-0-323-88427-3.00022-2>.
- de Klerk, D. and Schwarz, C. 2023. Simplified Approach to Understanding, Evaluating, and Parameterizing the NRTL Model for the Description of Binary VLE: Ttt-VLE Approach. *Industrial & Engineering Chemistry Research* **62** (27): 10629–10643. <https://doi.org/10.1021/acs.iecr.3c00576>.
- Demirel, E. and Ayas, N. 2017. Thermodynamic modeling of the water-gas shift reaction in supercritical water for hydrogen production. *Theoretical Foundations of Chemical Engineering* **51** (1): 76–87. <https://doi.org/10.1134/S0040579517010067>.
- Ganjdanesh, R., Rezaveisi, M., Pope, G. et al 2016. Treatment of Condensate and Water Blocks in Hydraulic-Fractured Shale-Gas/Condensate Reservoirs. *SPE Journal* **21** (2): 665–674. <https://doi.org/10.2118/175145-pa>.
- Henley, E., Seader, J., and Roper, D. 2011. Separation process principles, 3rd ed.: Hoboken, N.J. Wiley.
- Hsieh, M. and Lin, S. 2012. Effect of mixing rule boundary conditions on high pressure (liquid+liquid) equilibrium prediction. *The Journal of Chemical Thermodynamics* **47**: 33–41. <https://doi.org/10.1016/j.jct.2011.09.023>.
- Kong, S., Feng, G., Liu, Y. et al 2021. Potential of dimethyl ether as an additive in CO₂ for shale oil recovery. *Fuel* **296**: 120643. <https://doi.org/10.1016/j.fuel.2021.120643>.
- Lee, Y., Lee, H., Jeong, M. et al 2021. Compositional Modeling of Dimethyl Ether-CO₂ Mixed Solvent for Enhanced Oil Recovery. *Applied Sciences* **11** (1): <https://doi.org/10.3390/app11010406>.

- Léonard, G., Giuliani, D., and Villarreal-Singer, D. 2016. Design and Evaluation of a High-Density Energy Storage Route with CO₂ Re-Use, Water Electrolysis and Methanol Synthesis. *Computer Aided Chemical Engineering, Elsevier*, **38**, (.), 1797–1802. <https://doi.org/10.1016/B978-0-444-63428-3.50304-0>.
- Leonzio, G. 2018. State of art and perspectives about the production of methanol, dimethyl ether and syngas by carbon dioxide hydrogenation. *Journal of CO₂ Utilization* **27**: 326–354. <https://doi.org/10.1016/j.jcou.2018.08.005>.
- Li, S., Peng, D., Feng, S. et al 2023. Dimethyl ether-steam assisted gravity drainage: Physical 2D heavy oil simulation. *Fuel* **342**: 127821. <https://doi.org/10.1016/j.fuel.2023.127821>.
- Lotfollahzade Moghaddam, A. and Hazlett, M. 2023. Methanol dehydration catalysts in direct and indirect dimethyl ether (DME) production and the beneficial role of DME in energy supply and environmental pollution. *Journal of Environmental Chemical Engineering* **11** (3): 110307. <https://doi.org/10.1016/j.jece.2023.110307>.
- Miller, B. 2011. Clean Coal Engineering Technology, 2nd ed.: Elsevier Inc.
- Moioli, S. and Pellegrini, L. 2016. Modeling the methyldiethanolamine-piperazine scrubbing system for CO₂ removal: Thermodynamic analysis. *Frontiers of Chemical Science and Engineering* **10** (1): 162–175. <https://doi.org/10.1007/s11705-016-1555-5>.
- Neshat, S., Okuno, R., and Pope, G. 2020. Simulation of solvent treatments for fluid blockage removal in tight formations using coupled three-phase flash and capillary pressure models. *Journal of Petroleum Science and Engineering* **195**: 107442. <https://doi.org/10.1016/j.petrol.2020.107442>.
- Nielsen, M. July 2020. What "is" an Equation of State?, *The Way Ahead*, <https://jpt.spe.org/twa/what-equation-state> accessed 23 May 2025.
- Patel, S. March 2023. Non-Ideality Through Fugacity and Activity, *Department of Chemistry and Biochemistry, University of Delaware*, https://www1.udel.edu/pchem/C443/PatelLectures2016/Non-Ideality_Lectures.pdf accessed 23 May 2025.
- Peinado, C., Liuzzi, D., Sluijter, S. et al 2024. Review and perspective: Next generation DME synthesis technologies for the energy transition. *Chemical Engineering Journal* **479**: 147494. <https://doi.org/10.1016/j.cej.2023.147494>.
- PubChem. 2024. Dimethyl Ether, *National Library of Medicine*, 2024, <https://pubchem.ncbi.nlm.nih.gov/compound/Dimethyl-Ether> accessed 24 May 2025.
- Qayyum, A., Ali, U., and Ramzan, N. 2024. Acid gas removal techniques for syngas, natural gas, and biogas clean up—a review. *Energy Sources, Part A: Recovery, Utilization, and Environmental Effects* **46** (1): 13136–13159. <https://doi.org/10.1080/15567036.2020.1800866>.
- Ray, M. 1995. Coulson and Richardson's Chemical Engineering Volume 3 (Chemical & Biochemical Reactors, and Process Control), 3rd Edition, by J.F. Richardson and D.G. Peacock (Eds), Pergamon Press, Oxford, UK (1994). 776 pages. ISBN 0-08-041002-2.
- Developments in Chemical Engineering and Mineral Processing **3** (1): 56–57. <https://doi.org/10.1002/apj.5500030108>.
- Salahudeen, N., Ahmed, O., Rasheed, A. et al 2024. Exergy analysis of direct method for conversion of natural gas to methanol. *Discover Chemical Engineering* **4** (1): 19. <https://doi.org/10.1007/s43938-024-00058-5>.
- Sheng, K., Okuno, R., and Wang, M. 2018. Dimethyl Ether as an Additive to Steam for Improved Steam-Assisted Gravity Drainage. *SPE Journal* **23** (4): 1201–1222. <https://doi.org/10.2118/184983-pa>.
- Wang, C., Su, Y., Wang, W. et al 2024. Experimental Study on Ethane-Dimethyl Ether Assisted CO₂ Displacement to Improve Crude Oil Recovery in Tight Reservoirs. *Energy & Fuels* **38** (14): 12546–12554. <https://doi.org/10.1021/acs.energyfuels.4c00675>.
- Whitson wiki. 2024. Peng-Robinson Equation of State, <https://wiki.whitson.com/eos/eosmodels/preos/> accessed 23 May 2025.
- Ying, W., Genbao, L., Wei, Z. et al 2008. Study on the application of DME/diesel blends in a diesel engine. *Fuel Processing Technology* **89** (12): 1272–1280. <https://doi.org/10.1016/j.fuproc.2008.05.023>.